

Understanding Analysis, 2nd Edition

Chapter 6 - Section 6.4 - Series of Functions

1. Supply the details for the proof of the Weierstrass M-Test (Corollary 6.4.5).

Solution: We have for every fixed x

$$0 \leq |f_n(x)| \leq M_n$$

and $\sum_{n=1}^{\infty} M_n$ converges. This satisfies the conditions for the Comparison Test, so $\sum_{n=1}^{\infty} |f_n(x)|$ converges.

Then by the Absolute Convergence Test, $\sum_{n=1}^{\infty} |f_n(x)|$ converges implies $\sum_{n=1}^{\infty} f_n(x)$ converges.

It remains to prove that the convergence is uniform. $\sum_{n=1}^{\infty} M_n$ converges so it satisfies the Cauchy Criterion for Series (Theorem 2.7.2): for any $\epsilon > 0$, there exists N such that for all $n > m \geq N$:

$$M_{m+1} + M_{m+2} + \dots + M_n < \epsilon$$

Reusing the same N , we can prove that $\sum_{n=1}^{\infty} f_n(x)$ satisfies the Cauchy Criterion for Uniform Convergence of Series (Theorem 6.4.4). Because $|f_n(x)| \leq M_n$, then

$$\begin{aligned} |f_{m+1}(x) + f_{m+2}(x) + \dots + f_n(x)| &\leq |f_{m+1}(x)| + |f_{m+2}(x)| + \dots + |f_n(x)| \\ &\leq M_{m+1} + M_{m+2} + \dots + M_n \\ &< \epsilon \end{aligned}$$

Since the same N works for any x (because the bound M_n is independent of x), the convergence is uniform.